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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech (CBCS) Degree Examinations

Supplementary Examination - December 2025

MA1002-1 – CALCULUS AND DIFFERENTIAL EQUATIONS

(For AD, AM, CB, CC, CS, IS, RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. If ϕ be the angle between the tangent and radius vector at any point on the curve, $r = f(\theta)$ then $\cot \phi$ is equals to:

A) $r \frac{dr}{d\theta}$

C) $r \frac{d\theta}{dr}$

B) $\frac{1}{r} \frac{dr}{d\theta}$

D) $\frac{1}{r} \frac{d\theta}{dr}$
2. The radius of curvature of the curve $y = e^x$ at the point $(0, 1)$ is:

A) $2\sqrt{2}$

C) 2

B) $\sqrt{2}$

D) $\frac{\sqrt{2}}{2}$
3. The derivative of $\text{arc } \frac{ds}{dy}$ for the curve $x = f(y)$ is:

A) $\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

C) $1 + \left(\frac{dx}{dy}\right)^2$

B) $1 + \left(\frac{dy}{dx}\right)^2$

D) $\sqrt{1 + \left(\frac{dx}{dy}\right)^2}$
4. Curvature of a straight line is :

A) ∞

C) 1

B) 0

D) 2
5. Slope of the curve $y^2 + 2x^2 = 6$ at $(1, 1)$ is:

A) 2

C) 3

B) 4

D) -2
6. If $f(x)$ is a differentiable function and $f(1) = f(4)$ then

A) $f'(c) = 0$ for some $c \in (1, 4)$

C) $f'(c) = 0$ for some $c \in (3, 4)$

B) $f'(c) = 0$ for some $c \in (1, 2)$

D) $f'(c) = 0$ for some $c \in (2, 4)$
7. If $f(x, y, z) = x^2 + xyz + z$, then the value of f_x at $(1, 1, 1)$

A) 0

C) -1

B) 1

D) 3
8. If $u = x - y$ and $v = x + y$, then the Jacobian $\frac{\partial(u,v)}{\partial(x,y)}$ is:

A) 2

C) -2

B) 3

D) 4

9. If $f(x, y) = 2xy + 7x^2$ then Hessian of f is
 A) $\begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix}$
 B) $\begin{bmatrix} 14 & 2 \\ 2 & 0 \end{bmatrix}$
 C) $\begin{bmatrix} 7 & 1 \\ 1 & 0 \end{bmatrix}$
 D) $\begin{bmatrix} 2 & 1 \\ 7 & 0 \end{bmatrix}$
10. If $\phi = x^2 + y^2 + z^2$ then $\nabla\phi$ at $(1, 1, 1)$ is
 A) $\hat{i} + 3\hat{j} + 5\hat{k}$
 B) $\hat{i} + \hat{j} + \hat{k}$
 C) $2\hat{i} + 2\hat{j} + 2\hat{k}$
 D) $\hat{i} + 2\hat{j} + \hat{k}$
11. A vector field with curl zero is called
 A) an irrotational vector field
 B) solenoidal vector field
 C) scalar field
 D) Convergent field
12. If $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$, the divergence of $\vec{F}$ is:
 A) 0
 B) 1
 C) 3
 D) 2
13. Which of the following equation is not a first order linear differential equation in y ?
 A) $\frac{dy}{dx} + x^2y = \cos^2 x$
 B) $\frac{dy}{dx} + y \sec^2 x = e^x$
 C) $\frac{dy}{dx} + y \sin^2 x = x^3$
 D) $\frac{dy}{dx} + x^2y^2 = \tan x$
14. If two roots of the auxiliary equation of a second order linear differential equation with constant coefficients are purely imaginary, then the complementary solution is of the form:
 A) $y_c = A \cos \beta x$
 B) $y_c = A \cos \beta x + B \sin \beta x$
 C) $y_c = A \sin \beta x$
 D) $y_c = e^{\alpha x}(A \cos \beta + B \sin \beta)$
15. The partial differential equation of the expression $z = ax - by$, where a and b are arbitrary constants, is:
 A) $z = px - qy$
 B) $z = px + qy$
 C) $z = p + q$
 D) $px + qy = 0$
16. The following second order partial differential equation is: $2 \frac{\partial^2 u}{\partial x^2} - x^2 \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$
 A) Parabolic
 B) Hyperbolic
 C) Elliptic
 D) Circular
17. Area of a region R in terms of polar co-ordinate is
 A) $\iint_R r \, dr \, d\theta$
 B) $\iint_R dr \, d\theta$
 C) $\iint_R r^2 \, dr \, d\theta$
 D) $\iint_R (r + 1) \, dr \, d\theta$
18. The value of $\int_0^1 \int_0^1 \int_0^1 2z \, dz \, dx \, dy$ is:
 A) 2
 B) 1
 C) $\frac{1}{8}$
 D) $\frac{1}{4}$
19. The value of $\left(\Gamma\left(\frac{1}{2}\right)\right)^2$ is:
 A) $\sqrt{\pi}$
 B) $\frac{\pi}{2}$
 C) π
 D) $\frac{\pi}{4}$
20. For any $m > 0$, $\beta(m, 1) =$
 A) 1
 B) m
 C) 0
 D) $\frac{1}{m}$

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I**

	Marks	BT*	CO*	PO*
1. a) If ρ_1 and ρ_2 are the radii of curvature at the extremities of any focal chord of the cardioid $r = a(1 + \cos \theta)$ which passes through the pole, show that $\rho_1^2 + \rho_2^2 = \frac{16a^2}{9}$.	6	L*3	1	1
b) If $u = f(2x - 3y, 3y - 4z, 4z - 2x)$, show that $\frac{1}{2} \frac{\partial u}{\partial x} + \frac{1}{3} \frac{\partial u}{\partial y} + \frac{1}{4} \frac{\partial u}{\partial z} = 0$.	6	L2	2	1
c) Obtain the Taylor's series expansion for the function $f(x) = \log x$ about the point $x = 1$, up to two non-vanishing terms.	4	L2	1	2
2. a) With usual notations, prove that $\tan \phi = r \frac{d\theta}{dr}$.	6	L2	1	1
b) A rectangular box open at the top is to have volume of 32 cubic ft. Find the dimensions of the box requiring least material for its construction.	6	L3	2	2
c) If $u = \log \left(\frac{x^4 + y^4}{x + y} \right)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3$.	4	L2	2	1
3. a) State and prove Cauchy's mean value theorem.	6	L2	1	1
b) If $u = \frac{yz}{x}$, $v = \frac{xz}{y}$ and $w = \frac{xy}{z}$, show that $\frac{\partial(u,v,w)}{\partial(x,y,z)} = 4$.	6	L3	2	1
c) Given $u = \sin \left(\frac{x}{y} \right)$, $\dot{x} = e^t$ and $y = t^2$, find $\frac{du}{dt}$ at $t = 1$.	4	L2	2	1

Unit – II

4. a) If $\vec{A} = x^2y \hat{i} - 2xz \hat{j} + 2yz \hat{k}$, find $\text{curl}(\text{curl } A)$ and $\text{grad}(\text{div } A)$.	6	L2	3	2
b) Solve the differential equation $(D^3 - 6D^2 + 11D - 6)y = e^{-2x} + e^{-3x}$.	6	L3	4	1
c) Show that the vector function $\vec{V} = (x + 2y + 4z)\hat{i} + (2x - 3y - z)\hat{j} + (4x - y + 2z)\hat{k}$ is irrotational.	4	L2	3	1
5. a) Solve the differential equation $y'' - 4y' + 4y = (e^x + 32 \sin 2x + 16x)$.	6	L2	4	1
b) Solve the differential equation $y^3 \frac{\partial z}{\partial x} + x^2 \frac{\partial z}{\partial y} = 0$ by using the method of separation of variables.	6	L2	4	1
c) Determine the constant a so that the vector function $A = (x^2z - 24xyz)\hat{i} - (12x^2z + 2xyz)\hat{j} + (2xy^2 + ayz^2)\hat{k}$ is solenoidal.	4	L3	3	2
6. a) Find the directional derivative of the function $f(x, y, z) = xy^3 + yz^3$ at $(2, -1, 1)$ in the direction of the vector $\hat{i} + 2\hat{j} + 2\hat{k}$.	6	L3	3	1
b) Form the partial differential equation by eliminating the arbitrary function $F(x^2 + 2yz, y^2 + 2xz) = 0$.	6	L2	4	1
c) Solve the differential equation $y'' + 4y = 0$, $y(0) = -3$, $y'(0) = 1$.	4	L2	4	2

Unit – III

7. a) With usual notations, prove that $\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$.	6	L2	5	1
b) Find by double integration, the area bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.	6	L3	5	1
c) Evaluate $\int_0^{\frac{\pi}{2}} \sin^7 \theta \cos^7 \theta d\theta$ in terms of gamma function.	4	L2	5	1
8. a) Prove that $\int_0^{\frac{\pi}{2}} \sqrt{\sin \theta} d\theta \times \int_0^{\frac{\pi}{2}} \frac{d\theta}{\sqrt{\sin \theta}} = \pi$.	6	L2	5	1
b) Evaluate $\int_0^{\infty} \int_0^x x e^{-\frac{x^2}{y}} dy dx$ by changing the order of integration.	6	L3	5	1
c) Evaluate $\int_0^1 \int_0^x \int_0^{xy} xyz dx dy dz$.	4	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
